



TalentMatch AI — Enterprise AI Recruitment Platform

AI-Powered Resume Screening using Semantic Search, OCR, Vector Embeddings & Skill Matching

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[GitHub Repository](#) · [Live Demo](#)

Problem Statement

Why traditional recruitment falls short

- Recruiters manually review hundreds of resumes — time-consuming and inconsistent.
- Legacy ATS relies on literal keyword matching and misses semantic relevance.
- Scanned PDFs and images often fail standard parsers, losing candidate data.
- Manual screening increases time-to-hire and introduces bias and variance.

Organizations need an intelligent platform that understands resume meaning — not just keywords.



Solution Overview

How TalentMatch AI solves the problem



Resume Upload



PDF Text Extraction



OCR Processing



AI Resume Parsing



Vector Embedding Generation



Semantic Search



Skill Matching



Hybrid Scoring



Candidate Ranking



Recruiter Dashboard



Analytics Dashboard

TalentMatch AI automates the recruitment workflow from resume upload to intelligent candidate ranking.

System Architecture



Modular pipeline design: extraction and parsing are separated from embedding and search to enable scalability, observability, and independent upgrades.

Technology Stack



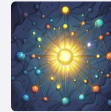
Frontend

React · TypeScript ·
Tailwind CSS ·
TanStack Router



Backend

Supabase ·
PostgreSQL ·
pgvector · Google
OAuth



AI

Gemini AI · OpenAI
text-embedding-3-
small · Cosine
similarity



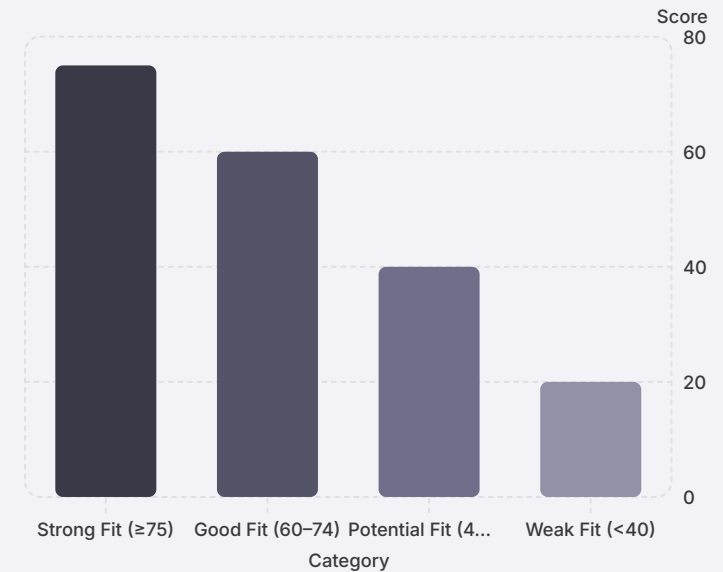
Processing

OCR · PDF Parser ·
Background
processing
(workers, queues)

Selected for scalability, vector search performance and secure user authentication.

AI Candidate Ranking — Scoring Methodology

We combine semantic similarity with explicit skill overlap to produce a robust ranking that balances meaning and concrete requirements.



Semantic Score

Cosine similarity between resume and job description embeddings.

Skill Score

Overlap between extracted candidate skills and job-required skills.

Final Score

Final = $0.8 \times \text{Semantic Score} + 0.2 \times \text{Skill Score}$

Bonus Rule

Complete skill coverage + semantic $\geq 60 \rightarrow$ additional boost.

Thresholds drive recruiter recommendations and shortlisting rules.

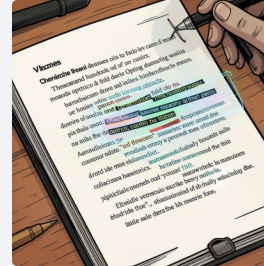
Key Features



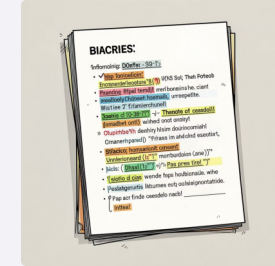
**Google
Authentication**



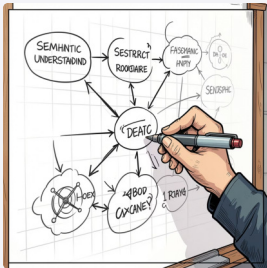
**Resume Upload (PDF
& image)**



OCR Support



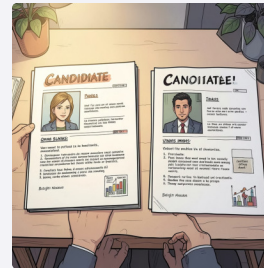
AI Resume Parsing



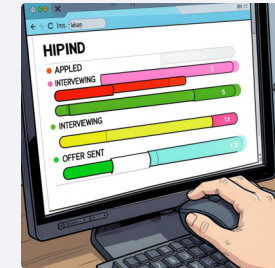
**Semantic Search &
Vector Embeddings**



**Skill Matching &
Hybrid Scoring**



**Candidate
Comparison & Notes**



**Candidate Status
Tracking**

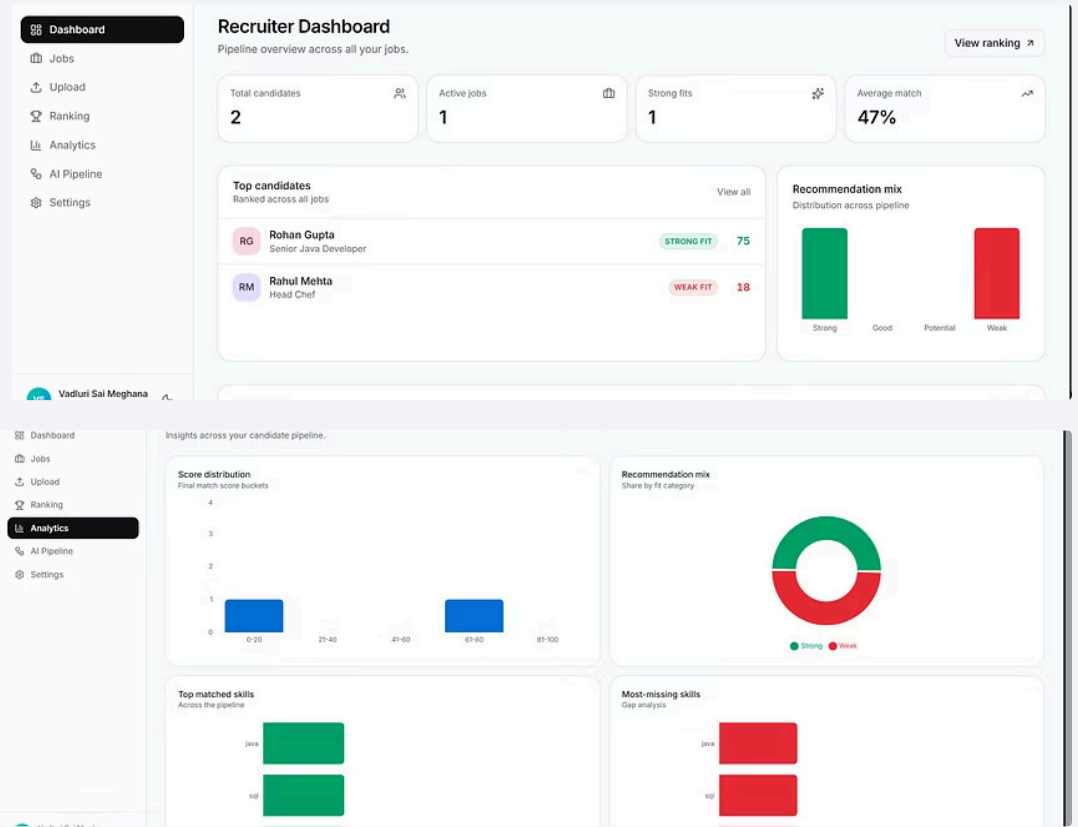
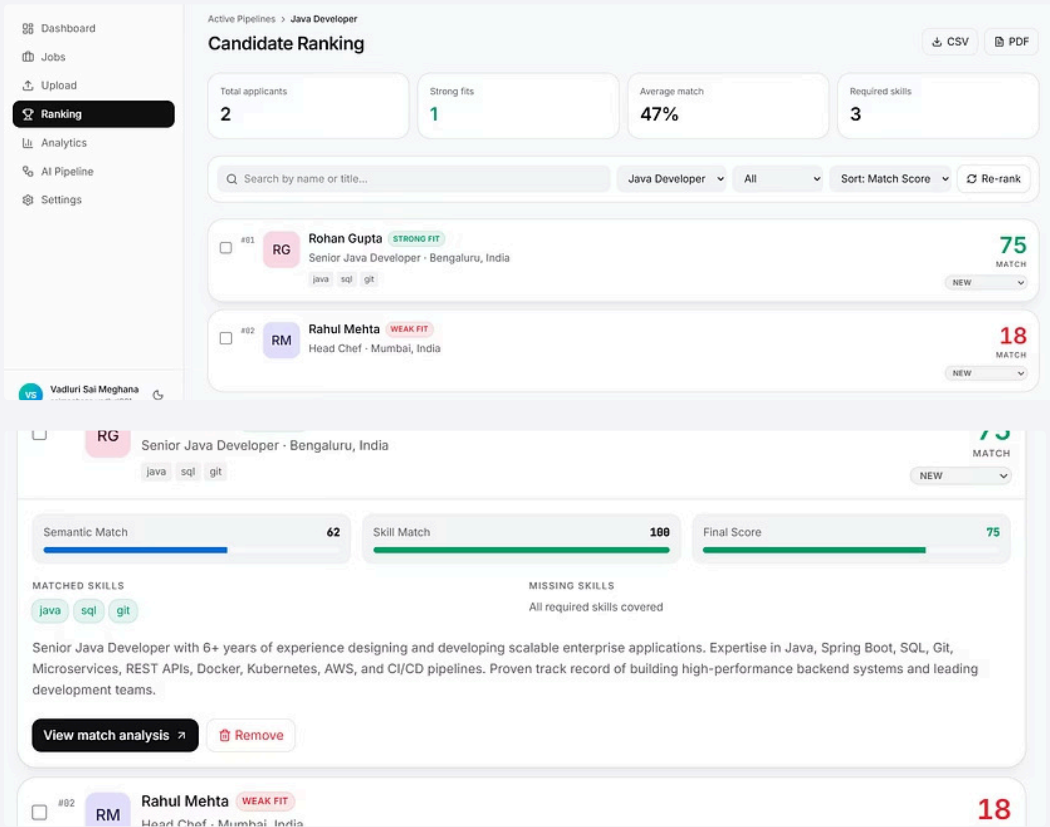


Analytics Dashboard



CSV / PDF Export

Website Demonstration (Screenshots)



Results — Performance & Ranking Example

Rank	Candidate	Score	Recommendation
1	Rohan Gupta	75	Strong Fit
2	Aarav Sharma	67	Good Fit
3	Priya Verma	60	Good Fit
4	Rahul Mehta	18	Weak Fit

High relevance yields top ranks; semantic and skill signals together distinguish borderline candidates from strong matches.

Design Decisions & Conclusion

OCR

Enables processing of scanned and image-based resumes reliably.

Vector Embeddings

Capture semantic meaning beyond keywords for better match quality.

Semantic Search

Improves recall and surface relevant candidates who use different phrasing.

Hybrid Scoring

Balances semantic relevance with explicit skill requirements.

The architecture increases accuracy while reducing manual screening time. Background processing and modular services ensure scalability for enterprise workloads.

Repository: [GitHub](#)

Live Demo: [ai-candidate-alchemy.lovable.app](#)

Thank you

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